Obesity Risk Prediction from Lifestyle Behaviors

Predicting overweight/obesity from everyday behaviors alone — no medical records required.

Data: CDC BRFSS 2022 · n = 329,126 **Model:** XGBoost + SMOTE **Deploy:** FastAPI on Render (live) **Context:** Master's ML final project

No medical record needed — daily behaviors alone reach F1 = 0.608; adding demographics reaches 0.680, nearly matching the 0.687 of a model with *every* chronic-disease diagnosis.

The marginal value of 7 chronic-disease history variables is < 1% F1 — evidence supporting behavior-first public-health intervention.

RESULTS — BINARY (NORMAL VS OVERWEIGHT+OBESE)

FEATURE SET	FEATS	F1
Full_25feat (+ chronic disease)	25	**0.687**
Behav_15feat (no disease hx)	15	**0.680**
Core_6feat (pure behavior)	6	**0.608**

KEY FINDINGS

1. **Chronic-disease history adds < 1% F1.** Behav_15 vs Full_25 differ by only 0.0067.
2. **The bottleneck is the Normal/Overweight boundary**, not the features — every group gains a uniform +0.19 F1 on the binary task.
3. **Cross-sectional ≠ longitudinal.** In 16-week simulations, the pure-behavior model tracks behavior change correctly, while the 15-feature model is anchored by static demographics — a property of survey data, and a key discussion point.

PIPELINE

Raw XPT 445k × 328 → ETL clean 329k × 26 → 27-model matrix → XGBoost + SMOTE → Live API Render

3 feature sets × 3 algorithms × 3 sampling strategies = 27 models, evaluated on weighted F1. Class imbalance handled with SMOTE; XGBoost won across all strategies.

LIVE API

`POST /predict/core6`	6 fields · F1 0.608
`POST /predict/behav15`	15 fields · F1 0.680
`GET /docs`	Swagger UI

Deployed on Render with an interactive demo presented in-browser, calling the live endpoint for real-time predictions.

TECH STACK

scikit-learn · XGBoost · SMOTE · pandas / numpy · FastAPI · Pydantic · Uvicorn · Render · matplotlib · GSAP